

Name:

Nick Milne

Date:

11/6/14

Mathematics Year 10 General Course

Test 3 Term 2, 2014

Topics – Trigonometry and Right Angled Triangles

<u>5</u>
30
= <u>17</u> %

Total Time: **40 minutes**Total Reading: **5 minutes**Total Working: **35 minutes**

Weighting: _____ % of the year.

CALCULATOR ASSUMED

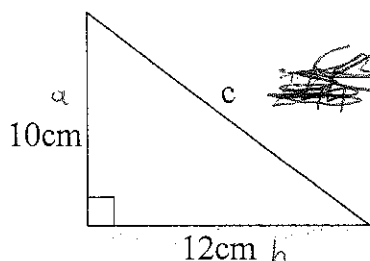
Time:	40 minutes	Marks for Section 2:	30
Reading:	5 minutes	Equipment Allowed:	½ page notes (A4 one side), Scientific calculator
Working:	35 minutes		

1. [2, 2, 2 marks]

Find the value of the missing side in each of the following triangles:

a.

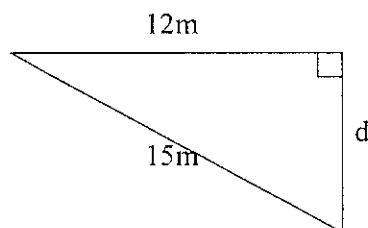
$$a^2 + b^2 = c^2$$



$$10^2 + 12^2 = 244$$

$$\sqrt{244} = 15.62 \text{ cm}$$

b.



Short side! so (take away).

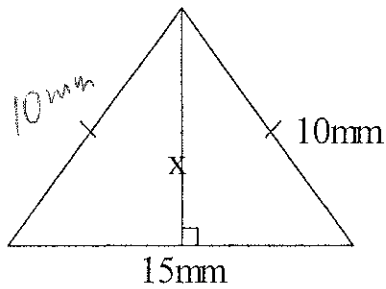
xx

$$12^2 + 15^2 = 369$$

$$\text{then } \sqrt{369} = 19.21$$

2.

c.



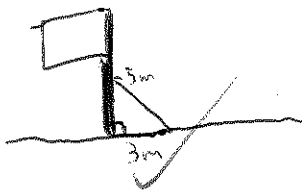
$$10 + 15 \div 2 = 17.5$$

x

2. [3 marks]

A support wire is attached 5 m up a flagpole. The other end is attached to the ground 3m from the base of the flagpole. How long is this wire?

(Hint: draw a quick sketch representing the situation.)

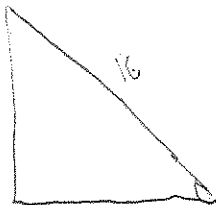


$$3^2 + 5^2 = 34^2$$

$$\sqrt{34} = 5.83m$$

3. [3 marks]

The length of the hypotenuse on an isosceles right-angled triangle is 16cm. What is the length of the other two sides? (Hint: draw a quick sketch representing the situation.)

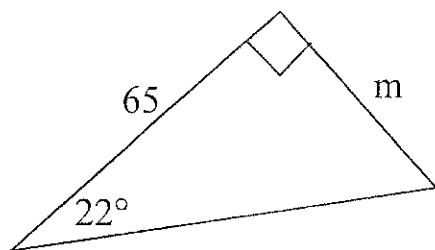


$$?^2 + ?^2 = 16^2$$

$$\sqrt{?} = 16$$

4. [2 marks]

Find the value of m , correct to three decimal places.



$$65 - \tan = 89.12$$

$\times$

5. [2 marks]

A 7.5m ladder rests against a wall so that its angle with the wall is 35° . How high up the wall does the ladder reach, correct to two decimal places?

$\times$

6. [2 marks]

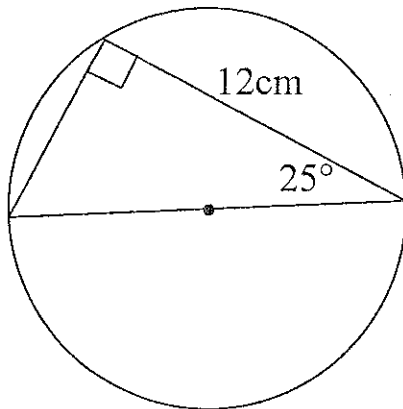
A ramp for disabled people runs from street level to the entrance foyer of a building. The entrance foyer is 0.95m above street level. How long is the ramp if it makes an angle of 12° to the horizontal, correct to two decimal places?

$\times$

$\checkmark$
0.

7. [2 marks]

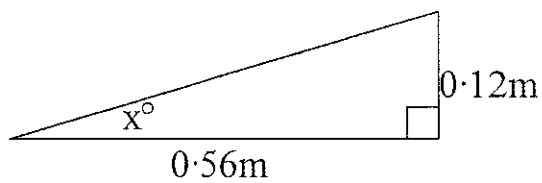
Find the radius of the circle shown below.



x

8. [2 marks]

Find the value of x in the following triangle.



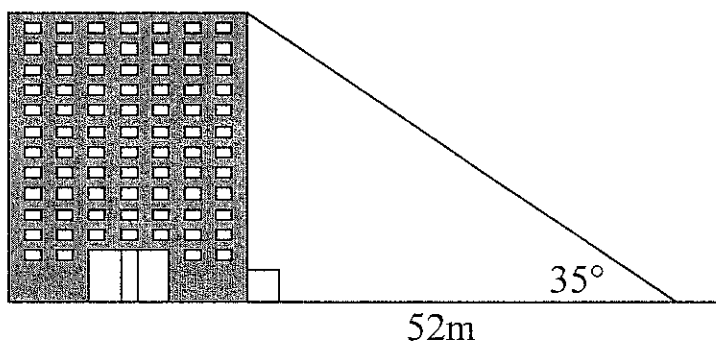
$$0.12^2 \div 0.56^2 = 0.056$$

$$\sqrt{0.056} = 0.236$$

?

9. [2 marks]

Find the height of the building below.



x

0

10. [3 marks]

A triangle has sides measuring 5cm, 7cm and 9cm. Is this triangle right-angled? Justify your answer by showing sufficient working/reasoning.

11. [3 marks]

A helicopter, flying at a constant altitude of 91m, sights the middle of a landing pad at an angle of depression of 16° . How far, in metres, must the helicopter travel to be directly over the landing spot?

10.

